

6. Acquisition B: edge-target and Z-step series

The edge series gives lateral (H, V) resolution and most of the boundary-bias data. The Z-step series gives depth resolution. Both use robot motion: lateral sub-pixel offsets for edges, and small Z steps for depth.

6.1 Edge series (B-HV, also feeds E)

1. Mount T3a (raised square) with $G = \text{GAP_SMALL_MM}$. Rotate it in its own plane by EDGE_SLANT_DEG , so all four edges are slanted relative to the pixel grid. The two near-vertical edges measure H resolution and the two near-horizontal edges measure V. Left and right edges have opposite occlusion geometry relative to the baseline, and so do top and bottom.
2. At each shape station ($\text{Z_SHAPE_STATION_STRIDE}$), centered and fronto-parallel: move, settle, and capture $\text{FRAMES_PER_EDGE_POSE}$ frames at the nominal pose.
3. At the same Z, capture $\text{PHASE_JITTER_POSES_EDGE}$ further poses. Each adds a logged random lateral offset, drawn uniformly over $\pm \text{PHASE_JITTER_SPAN_PX}/2$ in both H and V. Capture $\text{FRAMES_PER_EDGE_POSE}$ frames at each. These poses sample every sub-pixel phase of each edge for the robot-stepped ESF and for E.
4. Repeat Steps 2–3 with $G = \text{GAP_LARGE_MM}$. Comparing the two step heights tests whether the normalized ESF depends on step height. If it does, the depth pipeline is nonlinear and resolution must be quoted together with its step height.
5. Repeat Steps 1–4 with T3b (square window). The window has the opposite edge polarity: front plate outside, back plate inside.

6.2 Z-step series (B-Z)

1. Mount T2 centered and fronto-parallel at each Z_0 of the reduced stations ($\text{Z_REDUCED_STATION_STRIDE}$). Approach every visit from below: overshoot by $\text{Z_STEP_APPROACH_OVERSHOOT_MM}$ and move up onto the pose, so that backlash and compliance do not enter the $A \rightarrow B$ difference. The read-back robot pose, carried into the camera frame through the registration, is the step truth; log it at $\text{POSE_LOG_RESOLUTION_MM}$ or finer.
2. Step ladder. For each δ in Z_STEP_LADDER_MM , alternate the target between Z_0 and $Z_0 + \delta$ for Z_STEP_REPEATS cycles (A, B, A, B, ...). Capture $\text{FRAMES_PER_ZSTEP_POSE}$ frames at each visit. Log the read-back pose at every visit; analysis uses the read-back displacement, not the

commanded δ , as the step truth, and reports the ratio of `ROBOT_REPEATABILITY_MM` to each rung as the truth uncertainty.

3. Fine staircase. Sweep from Z_0 to $Z_0 + Z_STAIRCASE_QUANTA \times$ the expected quantum δZ_q (from Section 2, using the Tier-A q until A gives a measured value). Use steps of $\delta Z_q / Z_STAIRCASE_SUBDIVISION$, with `Z_STAIRCASE_FRAMES` frames per step and the read-back pose logged at each. This reveals whether the output moves in quantized steps, or is smoothed by the sensor's interpolation.
4. Blank windows. No extra capture is needed: the Z_0 frames from Step 2 serve as the no-step reference for the B-Z false-alarm threshold.